

Energy

Novel Prediction Modeling of Flat Plate Solar Collector using Kolmogorov–Arnold Networks (KAN) and Multi-Objective Performance Optimization via NSGA-II --Manuscript Draft--

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Abstract:	This research presents a novel framework combining CFD, interpretable machine learning, and multi-objective optimization (MOO). A high-fidelity dataset from 1000 CFD simulations covering eight design parameters including tube position, absorber plate thickness, tube diameter, tube length, tube thickness, mass flowrate, inlet temperature, and solar radiation was generated to assess thermal and hydraulic performance. The Kolmogorov-Arnold Network (KAN), a neural network architecture with learnable activation functions enabling enhanced interpretability, was implemented and benchmarked against the Multilayer Perceptron (MLP). KAN outperformed MLP with R2 of 0.992 and 0.993 for outlet temperature and pressure drop respectively, and reduced root mean squared errors by 64% and 44%. Pruned KAN models yielded explicit symbolic regression formulas. Validation against external study data showed the symbolic regression formulas achieved mean absolute errors of 0.34% in outlet temperature prediction. MOO using NSGA-II with the KAN surrogate model identified a Pareto front balancing heat transfer rate maximization and pressure drop minimization over six decision variables. For instance, a notable knee point was found at 140 W heat transfer rate and 2.5 Pa pressure drop for a single tube FPSC with tube length 1.25m, tube diameter 14.7mm, flowrate 0.001kgs-1, tube position 0.51, tube thickness 0.5mm, and plate thickness 1.74mm. This integrated methodology combining CFD data, interpretable machine learning, symbolic regression, and MOO offers a transparent and efficient framework for practical FPSC design optimization in renewable energy systems.